

1. Administrative & Study Identification

Full Study Title: The Prevalence, Disease Burden and Prognosis of COPD in Patients With Cardiovascular Diseases **Short Title/**
Acronym: PRECEDE **Protocol Number:** D2287L00041 **NCT Number:** NCT06909773 **Sponsor Name:** AstraZeneca **Investigational Product:** Cardiopulmonary Co-Management and Disease Education **Phase of Development:** Not Applicable **Medical Monitor:** AstraZeneca Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To describe the prevalence of COPD in subjects aged 40 years or older with 3 types of cardiovascular diseases (CVD): coronary heart disease (CHD), atrial fibrillation (AF), and chronic heart failure (CHF). **Secondary Objectives:** To observe the effect of cardiopulmonary co-management on the short-term prognosis of subjects with CVD and COPD, including hospitalization/readmission rates, and to evaluate changes in quality of life, symptoms, and activity tolerance (measured via SAQ and AFEQT scores).

2.2 Background & Rationale To address the disease burden of unrecognized and unmanaged Chronic Obstructive Pulmonary Disease (COPD) in patients with Cardiovascular Diseases (CVD). Having both conditions significantly impacts patient health outcomes. This study implements a diagnostic and co-management strategy to determine COPD prevalence among this population and to observe whether collaborative care between cardiologists and pulmonologists improves short-term prognosis and quality of life.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional / Diagnostic **Control Type:** Single-arm (Single Group Assignment) **Blinding:** Open-label **Randomization:** N/A

3.2 Planned Enrollment & Duration Target \$N\$: 3,000 participants **Number of Sites:** 69 locations **Estimated Study Start:** 28-Nov-2025 **Estimated Primary Completion:** 28-Oct-2026 **Estimated Study Completion:** 31-Dec-2026

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Give signed written informed consent to participate. 2. At least 40 years of age at the baseline visit. 3. Previous or newly diagnosed with at least one of 3 types of CVD: Coronary heart disease (CHD), Atrial fibrillation (AF), or Chronic heart failure (CHF). 4. Subjects have no absolute contraindications to spirometry testing and have the cognitive ability to conduct questionnaires.

4.2 Exclusion Criteria 1. Significant diseases or conditions, which, in the opinion of the investigator, may put the subject at risk or influence the results of the study. 2. Women who are pregnant, lactating, planning to become pregnant, or of childbearing potential not using an acceptable method of contraception. 3. Subjects who are not able to provide written informed consent. 4. Treatment with an investigational study drug or device in another clinical study within the last 30 days or five half-lives prior to the baseline visit, whichever is longer.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Cardiopulmonary co-management where cardiologists and pulmonologists work together to provide treatment management. Disease education is provided to patients every 4 weeks. **Route of Administration:** N/A (Healthcare management pathway and education). **Duration of Treatment:** Ongoing through the short-term follow-up period until the end of the study.

5.2 Concomitant & Prohibited Therapy Permitted: Standard of care medications and therapies for the management of COPD, CHD, AF, and CHF. **Prohibited:** Investigational study drugs or devices from other clinical trials within the last 30 days.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening/Baseline	Day 0	Informed Consent, Medical History, Spirometry testing, Baseline Questionnaires (SAQ, AFEQT), COPD/CVD Diagnostic Evaluation
Interim	Every 4 Weeks	Cardiopulmonary co-management follow-up, ongoing patient disease education by cardiologists and pulmonologists
End of Study	Final Visit	Assessment of hospitalization/readmission rates, final questionnaire scoring, exit evaluation

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: The number and percentage of COPD diagnoses in subjects with CVD (CHD, AF, CHF) to describe

the prevalence of COPD in the targeted population. **Measurement Method:** Clinical assessment and spirometry testing.

7.2 Secondary & Exploratory Endpoints * The hospitalization/readmission rate in subjects with CHD, AF, and CHF concurrent with COPD during the implementation of cardiopulmonary co-management. * Change from baseline in SAQ (Seattle Angina Questionnaire) score in subjects with CHD and COPD. * Change from baseline in AFEQT (Atrial Fibrillation Effect on Quality-of-life) score in subjects with AF and COPD. * Proportion of newly diagnosed COPD patients and patients identified as high risk for COPD at baseline.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection via regular 4-week follow-up patient visits. **Serious Adverse Events (SAEs):** Routine reporting timeline, typically within 24 hours of site awareness. **Data Monitoring Committee (DMC):** Internal safety and data monitoring committee managed by the sponsor (AstraZeneca).

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Full Analysis Set (FAS) for the evaluation of COPD prevalence and Intent-to-Treat (ITT) for evaluating secondary prognostic outcomes. **Sample Size Justification:** \$N = 3,000\$ participants, powered appropriately to determine an accurate epidemiological prevalence of COPD in the specific CVD sub-populations and track hospital readmission rates. **Handling of Missing Data:** Last Observation Carried Forward (LOCF) and/or Multiple Imputation for missing quality of life questionnaire items (SAQ, AFEQT).

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Investigators will receive specific COPD and CVD disease education according to Chinese Guidelines approximately 4 weeks before the first subject is enrolled to ensure quality and protocol adherence. **Audit Trail:** Standardized electronic Case Report Form (eCRF) data entry for spirometry results, clinical outcomes, and data clarification forms.

11. Study Identifiers & Governance

Primary ID: D2287L00041 **Registry Number:** NCT06909773 **Sponsor/Lead Organization:** AstraZeneca **Study Title:** The Prevalence, Disease

Burden and Prognosis of COPD in Patients With Cardiovascular Diseases **Official Regulatory Title:** The prevalence, disease burden and prognosis of COPD in patients with cardiovascular diseases - PRECEDE

12. Clinical Framework (COPD & CVD Specific)

Primary Condition: COPD in patients with Cardiovascular Diseases (CHD, AF, CHF) **Diagnosis Threshold:** Subjects ≥ 40 years old with confirmed CVD and spirometry-confirmed COPD **Medical Methodology:** Multidisciplinary Cardiopulmonary Co-Management **Optimization Target:** Improve short-term prognosis and reduce hospitalization/readmission rates

13. Study Procedures & Methodology

Management Pathway: Collaborative treatment management provided by cardiologists and pulmonologists **Education Protocol (HCP):** COPD and CVD disease education according to Guidelines ~4 weeks before the first subject is enrolled **Education Protocol (Patient):** Disease education on respective disease areas provided every 4 weeks **Quality Audit Mechanism:** Tracking of SAQ / AFEQT scores, and continuous monitoring of hospital readmissions

14. Observation & Follow-Up Schedule

Total Duration: Short-term prognosis follow-up through study completion **Onsite Visit Cadence:** Baseline, regular intervals, and End of Study **Remote Monitoring Frequency:** Follow-up management every 4 weeks **Checklist Review Interval:** Questionnaires evaluated at baseline and designated follow-up assessments

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	____	[DD-MMM-YYYY]				